

Devin Zarate

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EDUCATION

University of California, Irvine

B.S. in Electrical Engineering (Specialization in Digital Signal Processing)

Expected Dec 2027

GPA: 3.7

Relevant Coursework: Electronics I–III, Electronics Laboratory, Continuous & Discrete-Time Signals & Systems, Digital Logic Design, Network Analysis, Engineering Electromagnetics, Engineering Probability

Las Positas College

Electrical Engineering UC Pathway, Associate of Science

Livermore, CA

Aug 2023 – Jun 2025

PROJECTS

THEME PARK ENGINEERING DESIGN – Toronto Metropolitan Thrill Design Competition

Irvine, CA

Controls Engineer

Mar 2026 – Present

- Designed a redundant ride safety architecture integrating Hall-effect, inductive proximity, load, pressure, and vehicle-tracking sensors with a central PLC for fault detection and automated Emergency Stop (E-Stop) control.
- Developed PLC ladder logic for the complete ride control system, implementing redundant sensor validation, dispatch and E-Stop interlocks, and synchronized control of audio, lighting, and show effects.

UCI SOLAR CAR – Solar Powered Racecar

Irvine, CA

Motor Engineer

Mar 2026 – Present

- Implemented a high-voltage precharge sequence to mitigate destructive inrush currents to the motor controller's capacitor bank, calculating optimal power resistor values and configuring precise contactor timing for safe system initialization.
- Designed and fabricated a custom PCB integrating MOSFET-based driver circuits to control DCDC, precharge, and 96V contactors, interfacing low-voltage control signals with the vehicle's high-voltage power system.

FUSION ENGINEERING PROJECT – Fire Extinguishing Rover

Irvine, CA

Controls Engineer

Nov 2025 – May 2026

- Designed a custom ESP32-S3 control PCB integrating five H-bridge motor drivers, current-sensing feedback, and 5V/3.3V power regulation for multi-motor and servo control.
- Developed ESP32-S3 embedded firmware in C++ using Bluepad32, translating wireless gamepad inputs into coordinated real-time speed and direction control of four DC motors, two servos, and a pump.

INVISIBLE FENCE – Spatial Audio Filtering Using LTI Systems

Irvine, CA

Digital Signal Processing Project

Sept 2025 – Dec 2025

- Designed an LTI digital filtering system in MATLAB to isolate desired audio signals and suppress interfering frequency components from nearby sound sources.
- Applied FFT-based spectral analysis and convolution to identify dominant noise frequencies, characterize filter frequency response, and evaluate noise suppression in the time and frequency domains.

ACTIVITIES & LEADERSHIP

FUSION (Filipinx Undergraduate Science/Engineers In an Organized Network)

Sept 2025 – June 2026

Third Year Class Representative

- Cultivated a stronger sense of community for 60+ students by consistently facilitating communication and organizing casual social gatherings, including study breaks and game nights.

SKILLS

Languages: Python, C, C++, Java, MATLAB

Embedded & Hardware: ESP32, Arduino, I2C, CAN, UART, PCB Design, Soldering, Oscilloscopes

EDA & Tools: Altium Designer, KiCad, LTspice, Bluepad32